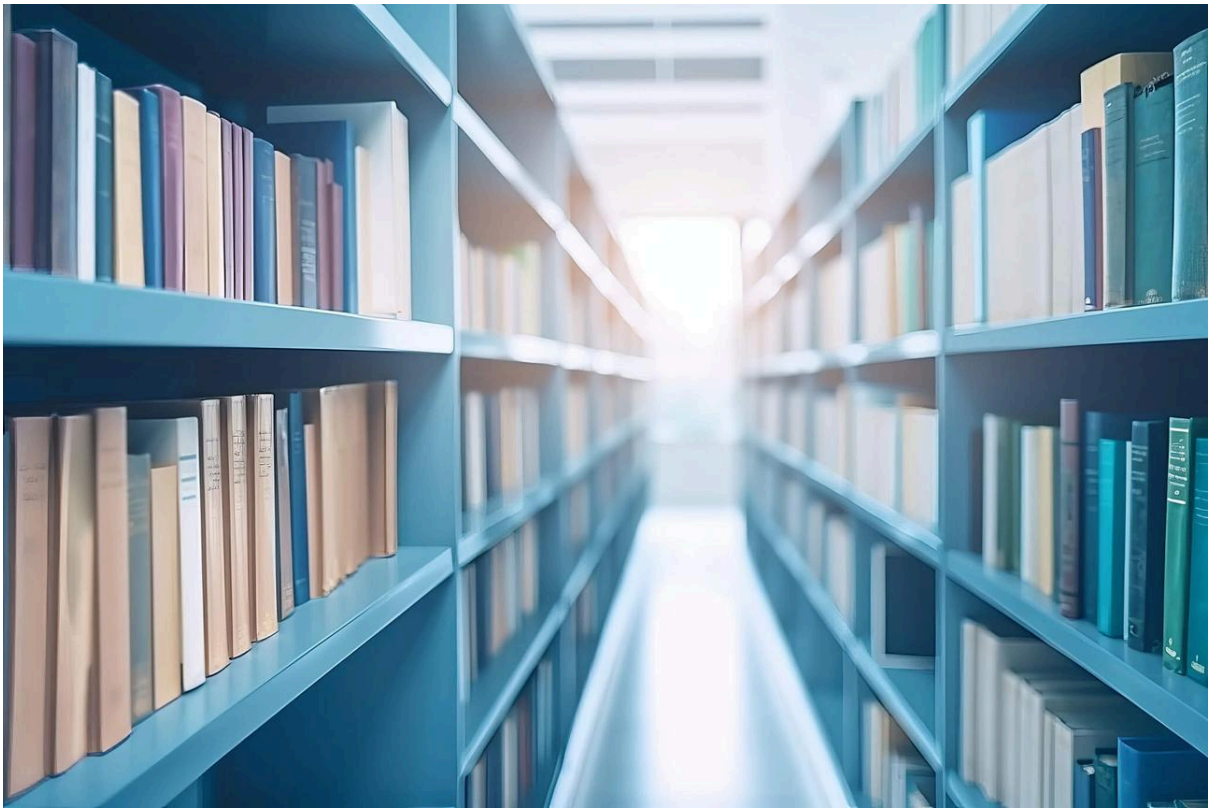
Digital Library Website – Explanation,Features and Development

1 Introduction

The Digital Library Management Website is a simple web-based application designed to display information about books in a library. It helps users view book categories, check whether books are available or issued, and browse a catalog of books.

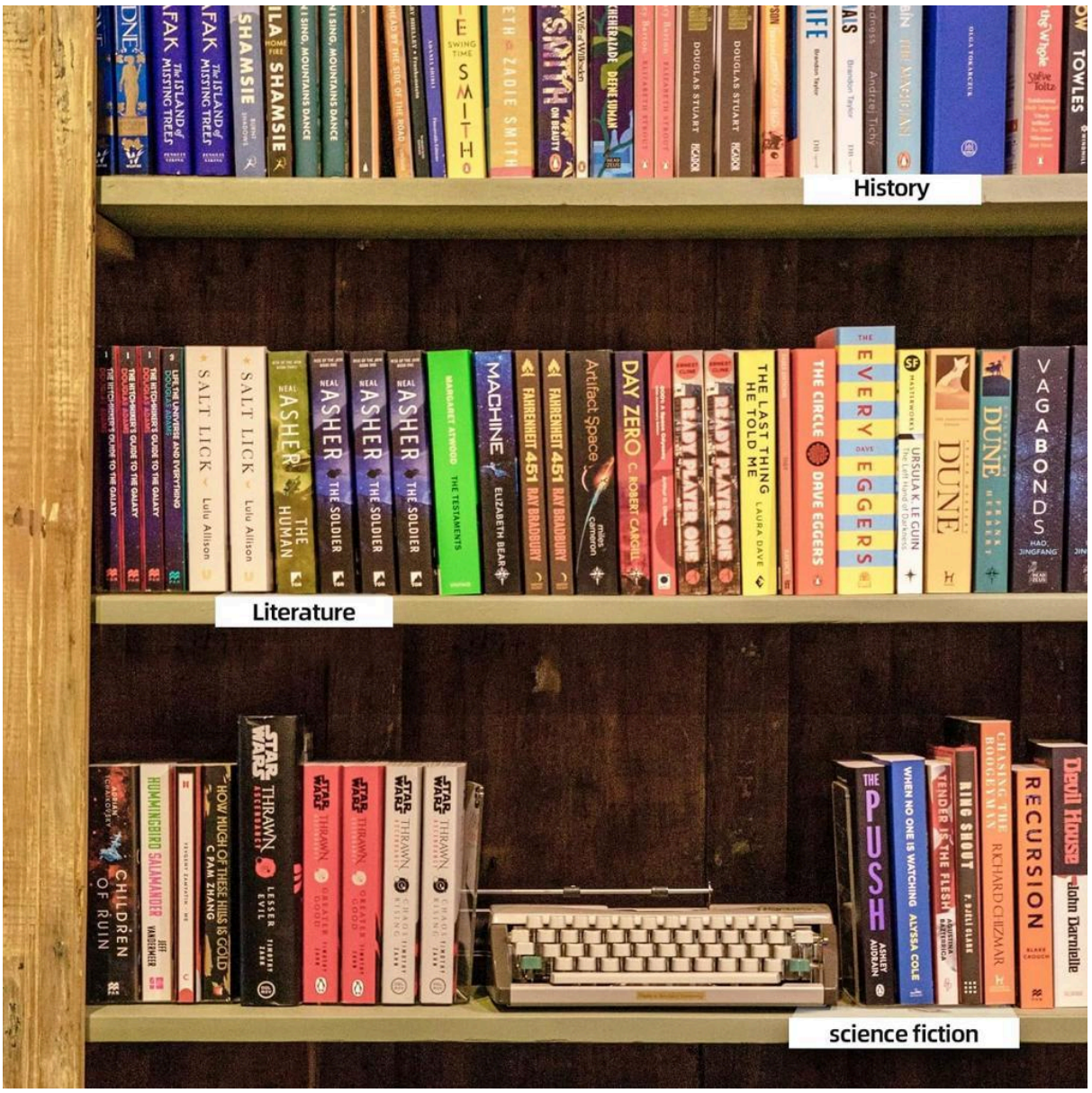
The website uses a blue-and-white color scheme to provide a clean, professional, and user-friendly interface. It also includes a responsive layout that adapts to smaller screens.

2 Features



1. Home Page

Displays the library title, a welcome message, and a navigation menu.



2. Book Categories

Organizes books into Computer Science, Science, Fiction, and Business categories.



3. Book Details

Displays book cover images, titles, authors, and availability status.



4. Search and Filter Interface

Provides a text input, category dropdown, and Search button. These are currently visual elements; searching and filtering are not implemented.



5. Book Catalog and Library Rules

Shows book IDs, titles, authors, categories, statuses, and rules for library users.

3 Explanation of HTML elements

HTML element	Purpose
`<!DOCTYPE html>`	Declares the document as HTML5.

`<html>`	Contains the webpage structure.
`<head>`	Contains document information and the title.
`<title>`	Sets the browser tab title to Digital Library.
`<style>`	Contains the CSS styling rules.
`<body>`	Contains the visible webpage content.
`<header>`	Displays the library name and tagline.
`<nav>`	Provides navigation links.
`<section>`	Groups related content such as categories, books, and rules.
`<div>`	Groups elements for styling and grid layouts.
`<img>`	Displays book cover images.
`<h1>`, `<h2>`, `<h3>`	Define headings at different levels.
`<p>`	Displays paragraphs and book information.
`<span>`	Displays availability or issued-status labels.

`<input>`	Provides a text field for entering search terms.
`<select>` and `<option>`	Create the category dropdown.
`<button>`	Displays the Search button.
`<table>`, `<tr>`, `<th>`, `<td>`	Display book information in tabular form.
`<ul>` and `<li>`	Display the library rules as a list.
`<footer>`	Displays copyright information.

Important HTML attributes

- `id` identifies a section, allowing navigation links such as `href="#books"` to jump to it.
- `class` applies common CSS styles to elements.
- `src` specifies the URL of a book cover image.
- `alt` provides alternative text if an image cannot be displayed.
- `href` specifies the destination of a navigation link.

4 Explanation of CSS

CSS property	Function
`margin: 0; padding: 0;`	Removes default browser spacing.
`box-sizing: border-box;`	Includes padding and borders in an element's specified dimensions.

`font-family: Arial, sans-serif;`	Sets the font style.
`background-color`	Sets the background color.
`color`	Sets the text color.
`text-align: center;`	Centers text and other inline content.
`padding`	Adds space inside an element.
`margin`	Adds space outside an element.
`display: grid;`	Arranges categories and book cards in a grid.
`grid-template-columns`	Defines the number and size of grid columns.
`gap`	Adds space between grid items.
`border-radius`	Rounds the corners of cards.
`box-shadow`	Adds a shadow to book cards.
`object-fit: cover;`	Fits book images into their designated dimensions.

`border-collapse: collapse;`	Combines adjacent table borders.
`tr:nth-child(even)`	Applies alternating background colors to even-numbered table rows.
`:hover`	Changes a link's color when the pointer moves over it.
`@media (max-width: 700px)`	Adjusts the layout for screens up to 700 pixels wide.

Responsive design

The media query changes the category and book grids to one column on smaller screens. It also stacks the search controls vertically, adjusts navigation spacing, and reduces table text and cell padding.

5 Development process

Step 1: Plan the website

Identify the required pages and features.

Step 2: Create the HTML structure

Add the header, navigation, sections, book cards, catalog, and footer.

Step 3: Apply CSS styling

Set colors, fonts, grids, spacing, and card styles.

Step 4: Add book information

Insert book images, authors, categories, and statuses.

Step 5: Test responsiveness

Check navigation, images, layout, and readability on different screens.

6 Advantages

- Simple and easy-to-understand interface.
- Organized presentation of books and categories.
- Clear distinction between available and issued books.
- Book information is easy to scan in a table.
- Responsive layout for mobile devices.
- Can serve as a foundation for a larger library system.

7 Limitations and future improvements

The current website is a static front-end design. It does not store library records or perform actual book-management operations.

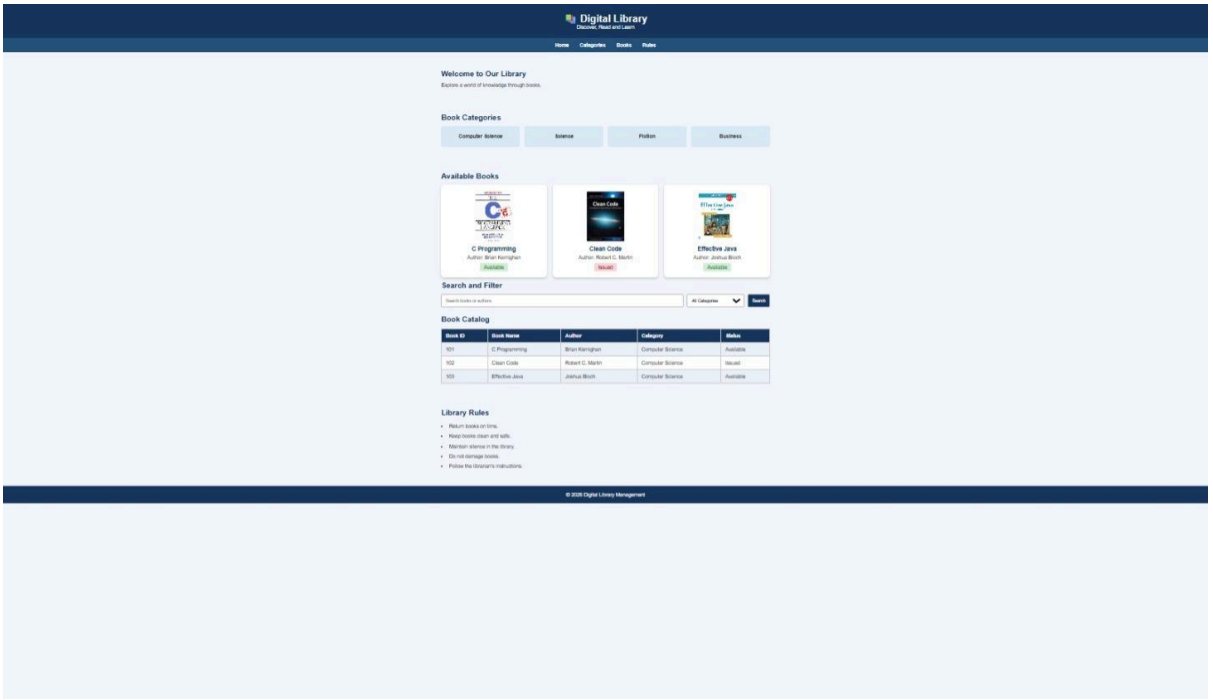
Possible improvements include:

- Implementing functional book search and category filtering using JavaScript.
- Connecting a database to store books and user records.
- Adding student and librarian login systems.
- Supporting book borrowing, returns, and reservations.
- Updating availability automatically.
- Adding form validation and a book-entry interface.

8 Conclusion

The Digital Library Management Website demonstrates how HTML and CSS can be used to create a structured, attractive, and responsive interface for presenting library information. It is suitable as a beginner-level web development project and can be extended into a functional library management application.

9 Output



Code

```
<!DOCTYPE html>
<html>
<head>
    <title>Digital Library</title>
</head>
<style>
    * {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    background-color: #f2f6fc;
    color: #333;
```

```
}

header {
    background-color: #17365d;
    color: white;
    text-align: center;
    padding: 25px;
}

nav {
    background-color: #24527a;
    padding: 15px;
    text-align: center;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 15px;
    font-weight: bold;
}

nav a:hover {
    color: yellow;
}

section {
    margin: 20px auto;
    padding: 25px;
    max-width: 1100px;
}

h2 {
```

```
    color: #17365d;
    margin-bottom: 15px;
    margin-top: 10px;
}

.grid {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 15px;
}

.category {
    background-color: #d9eaf7;
    padding: 25px 10px;
    text-align: center;
    border-radius: 8px;
    font-weight: bold;
}

.book-grid {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 20px;
}

.book {
    background-color: white;
    padding: 20px;
    text-align: center;
    border-radius: 10px;
    box-shadow: 0 3px 8px #ccc;
}
```

```
.book img {
    width: 120px;
    height: 160px;
    object-fit: cover;
    margin-bottom: 10px;
}

.book h3 {
    color: #17365d;
    margin-bottom: 8px;
}

.available {
    display: inline-block;
    background-color: #d4edda;
    color: green;
    padding: 5px 12px;
    margin-top: 10px;
    border-radius: 5px;
}

.issued {
    display: inline-block;
    background-color: #f8d7da;
    color: darkred;
    padding: 5px 12px;
    margin-top: 10px;
    border-radius: 5px;
}

.search {
    display: flex;
    gap: 10px;
```

```
    margin: 15px 0 25px;
}

input, select, button {
    padding: 12px;
    border: 1px solid #bbb;
    border-radius: 5px;
}

input {
    flex: 1;
}

button {
    background-color: #17365d;
    color: white;
    cursor: pointer;
}

button:hover {
    background-color: #24527a;
}

table {
    width: 100%;
    border-collapse: collapse;
    background-color: white;
    margin-top: 15px;
}

th, td {
    border: 1px solid #ccc;
    padding: 12px;
```

```
    text-align: left;
}

th {
    background-color: #17365d;
    color: white;
}

tr:nth-child(even) {
    background-color: #eaf1f8;
}

ul {
    padding-left: 25px;
    line-height: 2;
}

footer {
    background-color: #17365d;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 20px;
}

@media (max-width: 700px) {
    .grid, .book-grid {
        grid-template-columns: 1fr;
    }

    .search {
        flex-direction: column;
    }
```

```
    nav a {
        display: inline-block;
        margin: 5px;
    }

    table {
        font-size: 12px;
    }

    th, td {
        padding: 6px;
    }
}
</style>
<body>

<header>
    <h1>📚 Digital Library</h1>
    <p>Discover, Read and Learn</p>
</header>

<nav>
    <a href="#home">Home</a>
    <a href="#categories">Categories</a>
    <a href="#books">Books</a>
    <a href="#rules">Rules</a>
</nav>

<section id="home">
    <h2>Welcome to Our Library</h2>
    <p>Explore a world of knowledge through books.</p>
</section>
```

```
<section id="categories">
    <h2>Book Categories</h2>
    <div class="grid">
        <div class="category">Computer Science</div>
        <div class="category">Science</div>
        <div class="category">Fiction</div>
        <div class="category">Business</div>
    </div>
</section>

<section id="books">
    <h2>Available Books</h2>

    <div class="book-grid">
        <div class="book">
            <img
src="https://covers.openlibrary.org/b/isbn/9780131103627-L.jpg
" alt="C Programming book">
            <h3>C Programming</h3>
            <p>Author: Brian Kernighan</p>
            <span class="available">Available</span>
        </div>

        <div class="book">
            <img
src="https://covers.openlibrary.org/b/isbn/9780132350884-L.jpg
" alt="Clean Code book">
            <h3>Clean Code</h3>
            <p>Author: Robert C. Martin</p>
            <span class="issued">Issued</span>
        </div>
```

```
        <div class="book">
            <img
src="https://covers.openlibrary.org/b/isbn/9780134685991-L.jpg
" alt="Effective Java book">
            <h3>Effective Java</h3>
            <p>Author: Joshua Bloch</p>
            <span class="available">Available</span>
        </div>
    </div>

    <h2>Search and Filter</h2>
    <div class="search">
        <input type="text" placeholder="Search books or
authors">
        <select>
            <option>All Categories</option>
            <option>Computer Science</option>
            <option>Science</option>
            <option>Fiction</option>
            <option>Business</option>
        </select>
        <button>Search</button>
    </div>

    <h2>Book Catalog</h2>
    <table>
        <tr>
            <th>Book ID</th>
            <th>Book Name</th>
            <th>Author</th>
            <th>Category</th>
            <th>Status</th>
        </tr>
```

```
        <tr>
            <td>101</td>
            <td>C Programming</td>
            <td>Brian Kernighan</td>
            <td>Computer Science</td>
            <td>Available</td>
        </tr>
        <tr>
            <td>102</td>
            <td>Clean Code</td>
            <td>Robert C. Martin</td>
            <td>Computer Science</td>
            <td>Issued</td>
        </tr>
        <tr>
            <td>103</td>
            <td>Effective Java</td>
            <td>Joshua Bloch</td>
            <td>Computer Science</td>
            <td>Available</td>
        </tr>
    </table>
</section>

<section id="rules">
    <h2>Library Rules</h2>
    <ul>
        <li>Return books on time.</li>
        <li>Keep books clean and safe.</li>
        <li>Maintain silence in the library.</li>
        <li>Do not damage books.</li>
        <li>Follow the librarian's instructions.</li>
    </ul>
```

```
</section>

<footer>
    <p>© 2026 Digital Library Management</p>
</footer>

</body>
</html>
```